

E - Reverse and Reverse

分值: 800 分

Problem Statement

For a permutation $p = (p_1, p_2, \dots, p_N)$ of $(1, 2, \dots, N)$ and a positive integer M , let $f(p, M)$ denote the answer to the following problem.

对于 $(1, 2, \dots, N)$ 的一个排列 $p = (p_1, p_2, \dots, p_N)$ 和一个正整数 M , 我们用 $f(p, M)$ 表示如下问题的答案。

Perform the following operation M times on p .

- Choose an integer i with $1 \leq i \leq N - 1$, and reverse each of $(p_1, p_2, \dots, p_i)$ and $(p_{i+1}, p_{i+2}, \dots, p_N)$. Formally, replace p with $(p_i, p_{i-1}, \dots, p_1, p_N, p_{N-1}, \dots, p_{i+1})$.

There are $(N - 1)^M$ possible sequences of operations. Find the sum, modulo 998244353, of “the number of inversions of p after M operations” over all such sequences.

对 p 执行 M 次以下操作。

- 选择满足 $1 \leq i \leq N - 1$ 的整数 i , 然后分别翻转 $(p_1, p_2, \dots, p_i)$ 和 $(p_{i+1}, p_{i+2}, \dots, p_N)$ 。形式化地, 将 p 替换为 $(p_i, p_{i-1}, \dots, p_1, p_N, p_{N-1}, \dots, p_{i+1})$ 。

总共有 $(N - 1)^M$ 种可能的操作序列。求所有这些序列对应的「执行 M 次操作后 p 的逆序对个数」之和, 结果对 998244353 取模。

You are given a permutation $P = (P_1, P_2, \dots, P_N)$ of $(1, 2, \dots, N)$. Process the following query Q times.

给定一个 $(1, 2, \dots, N)$ 的排列 $P = (P_1, P_2, \dots, P_N)$, 你需要处理 Q 次询问。

- You are given an integer x with $1 \leq x \leq N - 1$ and a positive integer K . Swap P_x and P_{x+1} . Then, find $f(P, K)$.
- 给定满足 $1 \leq x \leq N - 1$ 的整数 x 和一个正整数 K 。交换 P_x 和 P_{x+1} , 然后求出 $f(P, K)$ 。

Constraints

- $2 \leq N \leq 2 \times 10^5$
 - $1 \leq Q \leq 2 \times 10^5$
 - P is a permutation of $(1, 2, \dots, N)$.
 - $1 \leq x \leq N - 1$
 - $1 \leq K \leq 10^9$
 - All input values are integers.
 - $2 \leq N \leq 2 \times 10^5$
 - $1 \leq Q \leq 2 \times 10^5$
 - P 是 $(1, 2, \dots, N)$ 的一个排列。
 - $1 \leq x \leq N - 1$
 - $1 \leq K \leq 10^9$
 - 所有输入值均为整数。
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Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

N Q
 $P_1 P_2 \dots P_N$
query₁
query₂
⋮
query _{Q}

Each query is given in the following format:

每个询问按照以下格式给出：

x K

Output

Output Q lines. The i -th line should contain the answer to query i .

输出 Q 行。第 i 行应包含第 query_i 个询问的答案。

Sample Input 1

```
3 2
1 3 2
1 1
2 1
```

Sample Output 1

```
4
4
```

For the first query, swapping P_1 and P_2 gives $P = (3, 1, 2)$. There are two possible sequences of operations as follows:

对于第一个询问，交换 P_1 和 P_2 后得到 $P = (3, 1, 2)$ 。共有两种可能的操作序列：

- Choose $i = 1$. P becomes $(3, 2, 1)$. The number of inversions is 3.
- Choose $i = 2$. P becomes $(1, 3, 2)$. The number of inversions is 1.
- 选择 $i = 1$, P 变为 $(3, 2, 1)$, 逆序对个数为 3。
- 选择 $i = 2$, P 变为 $(1, 3, 2)$, 逆序对个数为 1。

Thus, the answer is $3 + 1 = 4$.

因此答案为 $3 + 1 = 4$ 。

For the second query, swapping P_2 and P_3 gives $P = (3, 2, 1)$. There are two possible sequences of operations as follows:

对于第二个询问，交换 P_2 和 P_3 后得到 $P = (3, 2, 1)$ 。共有两种可能的操作序列：

- Choose $i = 1$. P becomes $(3, 1, 2)$. The number of inversions is 2.

- Choose $i = 2$. P becomes $(2, 3, 1)$. The number of inversions is 2.
- 选择 $i = 1$, P 变为 $(3, 1, 2)$, 逆序对个数为 2。
- 选择 $i = 2$, P 变为 $(2, 3, 1)$, 逆序对个数为 2。

Thus, the answer is $2 + 2 = 4$.

因此答案为 $2 + 2 = 4$ 。

Sample Input 2

```
4 4
3 2 4 1
2 1
2 2
3 3
1 4
```

Sample Output 2

```
11
28
67
242
```

Sample Input 3

```
10 7
7 9 3 10 5 2 4 6 8 1
2 29
1 86
3 30
8 64
1 24
1 9
5 55
```

Sample Output 3

29362950
633265500
847469581
741165544
385334408
653522086
169485402